

SQL Project: E-commerce Data Analysis

Project Overview

This project involves analyzing e-commerce data using SQL to extract meaningful insights related to customers, products, and sales performance.

Key Concepts Used

Joins, Aggregations, GROUP BY, Window Functions, CTEs, Date Functions

Key Business Queries & Solutions

1. Find Highest Marks

```
SELECT name FROM student WHERE marks = (SELECT MAX(marks) FROM student);
```

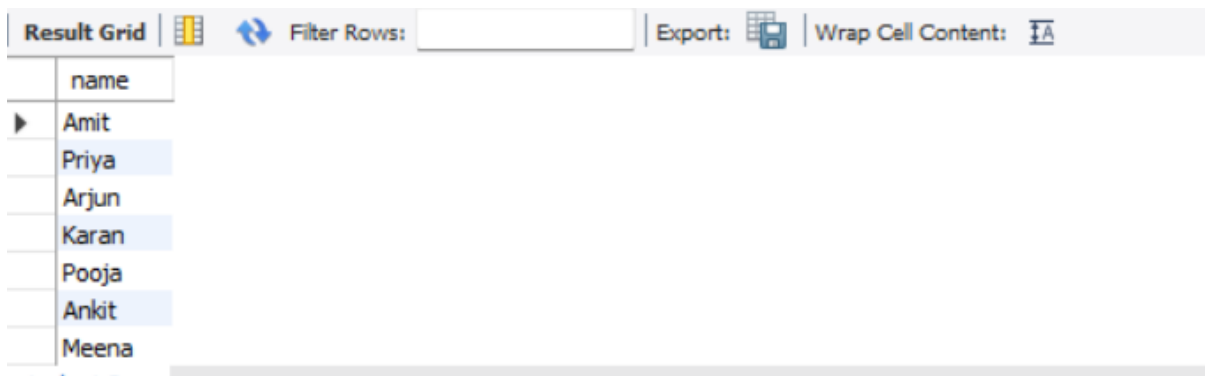


The screenshot shows a SQL query result grid. The toolbar at the top includes 'Result Grid', a grid icon, a refresh icon, 'Filter Rows:' with an input field, 'Export:' with a download icon, and 'Wrap Cell Content:' with a text icon. The table has two columns: 'name' and 'marks'. The first row shows the name 'Karan'.

	name
▶	Karan

2. Students Above Average Marks

```
SELECT name FROM student WHERE marks > (SELECT AVG(marks) FROM student);
```



The screenshot shows a SQL query result grid. The toolbar at the top includes 'Result Grid', a grid icon, a refresh icon, 'Filter Rows:' with an input field, 'Export:' with a download icon, and 'Wrap Cell Content:' with a text icon. The table has two columns: 'name' and 'marks'. The first row shows the name 'Amit'. The second row shows the name 'Priya'. The third row shows the name 'Arjun'. The fourth row shows the name 'Karan'. The fifth row shows the name 'Pooja'. The sixth row shows the name 'Ankit'. The seventh row shows the name 'Meena'.

	name
▶	Amit
	Priya
	Arjun
	Karan
	Pooja
	Ankit
	Meena

2. Female Customers who purchased Accessories

```
With CTE as (SELECT o.customer_id , p.category from Orders o  
LEFT JOIN Products p  
on o.product_id = p.product_id )
```

```

SELECT DISTINCT c.customer_name from CTE t
LEFT JOIN Customers c
on t.customer_id = c.customer_id
WHERE c.gender = 'Female'
AND t.category = 'Accessories'

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	customer_name			
▶	Alice			
	Carol			
	Grace			

4. Monthly Revenue

```

SELECT MONTH(STR_TO_DATE(order_date,'%d-%m-%Y')) AS month, SUM(p.price *
o.quantity) AS revenue FROM orders o
JOIN products p
ON o.product_id = p.product_id
GROUP BY month;

```

	month	revenue
▶	5	150
	6	430
	7	900
	8	1970

5. Month-over-Month Growth

```

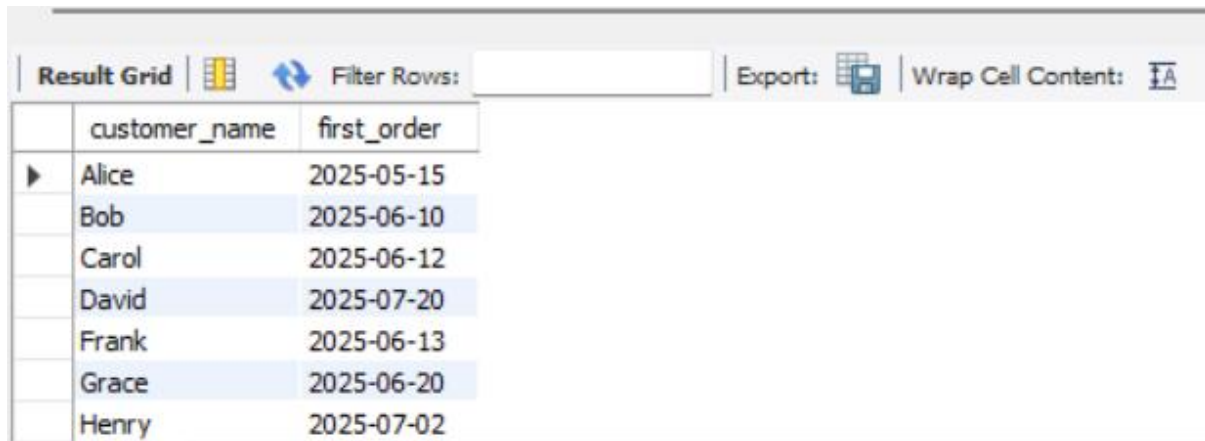
WITH cte AS ( SELECT MONTH(STR_TO_DATE(order_date,'%d-%m-%Y')) AS month,
SUM(p.price *o.quantity) AS revenue FROM orders o
JOIN products p
ON o.product_id = p.product_id
GROUP BY month)
SELECT month, ROUND((revenue - LAG(revenue) OVER (ORDER BY month))
/LAG(revenue) OVER (ORDER BY month) * 100,2) AS MOM FROM cte;

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	month	MOM		
▶	5	NULL		
	6	186.67		
	7	109.30		
	8	118.89		

6. First Order of Each Customer

```
SELECT c.customer_name, MIN(STR_TO_DATE(o.order_date,'%d-%m-%Y')) AS  
first_order FROM orders o  
JOIN customers c  
ON o.customer_id = c.customer_id  
GROUP BY c.customer_name;
```

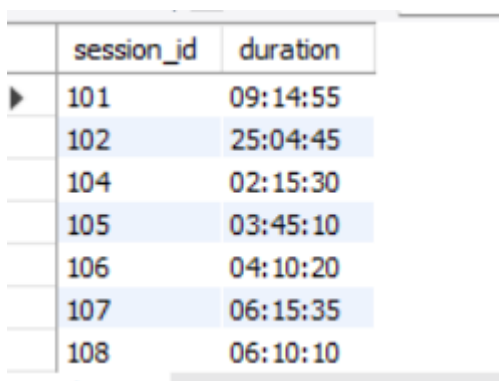


The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'customer_name' and 'first_order'. The data is as follows:

customer_name	first_order
Alice	2025-05-15
Bob	2025-06-10
Carol	2025-06-12
David	2025-07-20
Frank	2025-06-13
Grace	2025-06-20
Henry	2025-07-02

7. Session Duration

```
WITH cte AS ( SELECT a.session_id, a.time AS login, b.time AS logout FROM sessions a  
JOIN sessions b  
ON a.session_id = b.session_id  
AND a.status='login' AND b.status='logout')  
  
SELECT session_id, SEC_TO_TIME(TIMESTAMPDIFF(SECOND, STR_TO_DATE(login,'%d-  
%m-%Y %H:%i:%s'), STR_TO_DATE(logout,'%d-%m-%Y %H:%i:%s'))) AS duration FROM  
cte;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid displays the results of a SQL query. The columns are 'session_id' and 'duration'. The data is as follows:

session_id	duration
101	09:14:55
102	25:04:45
104	02:15:30
105	03:45:10
106	04:10:20
107	06:15:35
108	06:10:10

Conclusion

This project demonstrates the use of SQL for data analysis, including data cleaning, aggregation, and deriving insights from structured datasets.